Time series: ARMA forecasting

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The homework is provided on the final slides

Forecasting AR(3) process

Exercise. Consider the following AR(3) process $x_t = 0.2x_{t-1} - 0.2x_{t-2} + 0.6x_{t-3} + \epsilon_t$. Compute predictors $\hat{x}_{11}, \hat{x}_{12}$ and $\hat{x}_{13}$ for a time series from this AR process with $x_1 = 3.1$, $x_2 = 0.93$, $x_3 = -2.3$, $x_4 = 3.2$, $x_5 = 1.3$, $x_6 = -2.5$, $x_7 = 1.6$, $x_8 = 3.0$, $x_9 = -3.5$, $x_{10} = -0.74$.

According to slide 10 of Lecture 12, we have

$$\hat{x}_{11} = 0.2x_{10} - 0.2x_9 + 0.6x_8 = 0.2 \cdot (-0.74) - 0.2 \cdot (-3.5) + 0.6 \cdot 3 = 2.35$$

$$\hat{x}_{12} = 0.2\hat{x}_{11} - 0.2x_{10} + 0.6x_9 = 0.2 \cdot 2.35 - 0.2 \cdot (-0.74) + 0.6 \cdot (-3.5) = -1.482$$

$$\hat{x}_{13} = 0.2\hat{x}_{12} - 0.2\hat{x}_{11} + 0.6x_{10} = 0.2 \cdot (-1.482) - 0.2 \cdot 2.35 + 0.6 \cdot (-0.74) = -1.21$$

MSE of predictors for AR(1)

Exercise. A company sells cars. Sales (in units of 100 cars) over the past 10 years from 2003 to 2012 are $x_1, \ldots, x_{10}$ and $x_{10} = 40.3$. The company believes that the model $x_t - 40 = 0.6(x_{t-1} - 40) + \epsilon_t$, where ϵ_t is white noise with variance 0.6, is appropriate. Find the forecasts for the number of sales in 2013 and 2014 and the prediction bounds.

Based on slide 4 of Lecture 12, we have

$\hat{x}_{11} = \mu + a(x_{10} - \mu) = 40 + 0.6(40.3 - 40) = 40.18$,

$\hat{x}_{12} = \mu + a^2(x_{10} - \mu) = 40 + 0.6^2(40.3 - 40) = 40.11$.

Then $MSE(\hat{x}_{11}) = \sigma^2 = 0.6$ and

$MSE(\hat{x}_{12}) = \sigma^2 \frac{1-a^4}{1-a^2} = 0.6 \cdot 1.36 = 0.816$.

The $(1-\alpha)$-confidence level prediction bounds:

$(40.18 - z_{1-\alpha/2}\sqrt{0.6}\ ,\ 40.18 + z_{1-\alpha/2}\sqrt{0.6}\ )$ for x_{11} [2013]

$(40.11 - z_{1-\alpha/2}\sqrt{0.816}, 40.11 + z_{1-\alpha/2}\sqrt{0.816})$ for x_{12} [2014]

Forecasting AR(2) process

Exercise. Assume that the average air temperature (degrees C) in some state of Brazil in successive months behaves like an AR(2) process $x_t = 30 + 1.45(x_{t-1} - 30) - 0.58(x_{t-2} - 30) + \epsilon_t$ where ϵ_t is white noise with variance 0.2. Assume that the observations from January to April 2013 are 26, 34, 25, 36. Find the predictions and prediction bounds for May, June, July.

We have $x_1 = 26$, $x_2 = 34$, $x_3 = 25$, $x_4 = 36$. Therefore,

$\hat{x}_5 = 30 + 1.45(x_4 - 30) - 0.58(x_3 - 30) =$
$30 + 1.45 \cdot 6 - 0.58(-5) = 41.6$ [for May]

$\hat{x}_6 = 30 + 1.45(\hat{x}_5 - 30) - 0.58(x_4 - 30) =$
$30 + 1.45 \cdot 11.6 - 0.58 \cdot 6 = 43.34$ [for June]

$\hat{x}_7 = 30 + 1.45(\hat{x}_6 - 30) - 0.58(\hat{x}_5 - 30) =$
$30 + 1.45 \cdot 13.34 - 0.58 \cdot 11.6 = 42.615$ [for July]

Confidence intervals for AR(2) process

Exercise. Compute the prediction bounds for May, June and July (for the exercise from the previous slide).

According to slide 8 of Lecture 12, we have to compute the MA representation of the AR(2) process $\frac{1}{a(B)} = \frac{1}{1-[1.45B-0.58B^2]}$

$= 1 + [1.45B - 0.58B^2] + [1.45B - 0.58B^2]^2 + \dots$

$= 1 + 1.45B + [-0.58 + 1.45^2]B^2 + \dots$

Put $c_0 = 1$, $c_1 = 1.45$ and $c_2 = 1.52$ into $\mathrm{MSE}(\hat{x}_{n+h}) = \sigma^2 \sum_{i=0}^{h-1} c_i^2$.

Therefore, the prediction bounds are

$$May : (41.6 - z_{1-\alpha/2}\sqrt{0.2}\ ,\ 41.6 + z_{1-\alpha/2}\sqrt{0.2})$$

$$June : (43.34 - z_{1-\alpha/2}\sqrt{0.2}\sqrt{1+1.45^2},\ 43.34 + z_{1-\alpha/2}\sqrt{0.2}\sqrt{1+1.45^2})$$

$$July : (42.615 - z_{1-\alpha/2}\sqrt{0.2}\sqrt{1+1.45^2+1.52^2},\ 42.615 + z_{1-\alpha/2}\sqrt{0.2}\sqrt{1+1.45^2+1.52^2})$$

ARIMA(p,d,q) process

A process x_t is called an **autoregressive integrated moving average** [ARIMA(p,d,q)] process if the process $(1-B)^d x_t$ is an ARMA(p,q) process, $p,d,q \in \{0,1,2,\ldots\}$.

By definition, we have that an ARIMA process x_t satisfies

$$a(B)(1-B)^d x_t = b(B)\epsilon_t.$$

Note that for a given time series $x_1,\ldots,x_n$, the process $\tilde{x}_t = (1-B)x_t$ is a sequence of differences, that is, $\tilde{x}_2 = x_2 - x_1$, $\tilde{x}_3 = x_3 - x_2$, $\tilde{x}_4 = x_4 - x_3$, $\tilde{x}_5 = x_5 - x_4, \ldots$

Notation: x_t is referred to as the original series and $\tilde{x}_t := (1-B)x_t$ is called a series of differences.

ARIMA(0,1,0) process

An ARIMA(0,1,0) process x_t satisfies

$$(1 - B)x_t = \epsilon_t.$$

Rewriting we have

$$x_t - x_{t-1} = \epsilon_t$$

or

$$x_t = x_{t-1} + \epsilon_t.$$

Thus, an ARIMA(0,1,0) process x_t is a random walk process, see slide 5 of Lecture 4.

Also, x_t is a limiting case of an AR(1) process $(1 - aB)x_t = \epsilon_t$ as $a \to 1$.

ARIMA(1,1,0) process

An ARIMA(1,1,0) process x_t satisfies

$$(1 - aB)(1 - B)x_t = \epsilon_t$$

where $a \in (-1, 1)$.
Rewriting, we have $(1 - B - aB + aB^2)x_t = \epsilon_t$ or

$$x_t = (1 + a)x_{t-1} - ax_{t-2} + \epsilon_t.$$

An ARIMA(1,1,0) process is often used for modeling stock prices.
Forecasting of a time series $x_1, \ldots, x_n$ from ARIMA(1,1,0)
$\hat{x}_{n+1} = (1 + a)x_n \ - ax_{n-1}$
$\hat{x}_{n+2} = (1 + a)\hat{x}_{n+1} - ax_n$
$\hat{x}_{n+3} = (1 + a)\hat{x}_{n+2} - a\hat{x}_{n+1}$
and so on.

Forecasting ARIMA(1,1,0) process

Exercise. Suppose that the daily gold price (in ASD per ounce) follows an ARIMA process $x_t - 1.9x_{t-1} + 0.9x_{t-2} = \epsilon_t$ and observations for the last 5 days are 560, 580, 640, 770, 800. Calculate forecasts for the next 3 days.

Note that the given equation can be written as $(1 - 0.9B)(1 - B)x_t = \epsilon_t$, so we have an ARIMA(1,1,0) equation. Since $x_4 = 770$ and $x_5 = 800$ we have

$$\hat{x}_6 = 1.9x_5 - 0.9x_4 = 1.9 \cdot 800 - 0.9 \cdot 770 = 827$$

$$\hat{x}_7 = 1.9\hat{x}_6 - 0.9x_5 = 1.9 \cdot 827 - 0.9 \cdot 800 = 851.3$$

$$\hat{x}_8 = 1.9\hat{x}_7 - 0.9\hat{x}_6 = 1.9 \cdot 851.3 - 0.9 \cdot 827 = 873.17$$

Estimating AR(2) process

Exercise. Suppose that we have a time series $x_1, x_2, \ldots, x_{1500}$, and we have chosen to estimate an AR(2) model, $x_t = a_1 x_{t-1} + a_2 x_{t-2} + \epsilon_t$. Suppose that the estimators of the covariance function are $\hat{R}(0) = 5.0$, $\hat{R}(1) = 0.0$, $\hat{R}(2) = 2.5$, $\hat{R}(3) = 0.0$, $\hat{R}(4) = 1.0$. Estimate the coefficients a_1, a_2, and the noise variance σ^2.

From slides 3 and 8 of Lecture 11, we have $\hat{\rho}(1) = \hat{R}(1)/\hat{R}(0) = 0$ and $\hat{\rho}(2) = 2.5/5 = 1/2$. Solving Yule-Walker equations

$$\begin{pmatrix} 1 & \hat{\rho}(1) \\ \hat{\rho}(1) & 1 \end{pmatrix} \begin{pmatrix} \hat{a}_1 \\ \hat{a}_2 \end{pmatrix} = \begin{pmatrix} \hat{\rho}(1) \\ \hat{\rho}(2) \end{pmatrix},$$

we obtain the estimators $\hat{a}_1 = 0$ and $\hat{a}_2 = 1/2$.
The estimator of the noise variance is
$\hat{\sigma}^2 = \hat{R}(0)(1 - \hat{a}_1\hat{\rho}(1) - \hat{a}_2\hat{\rho}(2)) = 5(1 - 0 - 1/2 \cdot 1/2) = 5 \cdot 3/4 = 15/4$.

Estimating MA(1) process

Ex. Let the time series 8,7,5,4,8,6,4 follow an MA(1) process $x_t - \mu = \epsilon_t + b\epsilon_{t-1}$. Find the estimator of b and σ^2.

First, we compute the estimator of the mean,
$\hat{\mu} = (8+7+5+4+8+6+4)/7 = 6$.
Computing $y_i = x_i - \hat{\mu}$, $i = 1, \ldots, 7$ we obtain
$(y_1, \ldots, y_7) = (2, 1, -1, -2, 2, 0, -2)$.
Then we obtain
$\hat{R}(0) = (2^2 + 1^2 + (-1)^2 + (-2)^2 + 2^2 + 0 + (-2)^2)/7 = 18/7$
$\hat{R}(1) =$
$(2\cdot 1 + 1(-1) + (-1)(-2) + (-2)2 + 2\cdot 0 + 0(-2))/7 = -1/7$
Now we have to solve $\frac{b}{1+b^2} = \hat{\rho}(1) = -1/18$. Rewriting we have $b^2 + 18b + 1 = 0$ which has 2 solutions -0.056 and -17.94. Choosing a solution whose absolute value is smaller than 1 (to provide invertibility), we have $\hat{b} = -0.056$. Also, $\hat{\sigma}^2 = \hat{R}(0)/(1 + \hat{b}_1^2) = 18/7/(1 + (-0.056)^2) = 2.563$.

Parameter estimation for AR(2) process

Suppose that observations $x_1, \ldots, x_{100}$ have a sample mean of 30.1, sample variance 1.5 and sample correlations $\hat{\rho}(1) = 0.9$ and $\hat{\rho}(2) = 0.7$. Estimate parameters of an AR(2) process.

Note that an AR(2) process is given by $x_t - \mu = a_1(x_{t-1} - \mu) + a_2(x_{t-2} - \mu) + \epsilon_t$, where ϵ_t is white noise with variance σ^2.
First we have that $\hat{\mu} = 30.1$. Then solving

$$\begin{pmatrix} 1 & \hat{\rho}(1) \\ \hat{\rho}(1) & 1 \end{pmatrix} \begin{pmatrix} \hat{a}_1 \\ \hat{a}_2 \end{pmatrix} = \begin{pmatrix} \hat{\rho}(1) \\ \hat{\rho}(2) \end{pmatrix},$$

we have $\hat{a}_1 = \hat{\rho}(1)\frac{1-\hat{\rho}(2)}{1-\hat{\rho}^2(1)} = 1.42$, $\hat{a}_2 = \frac{\hat{\rho}(2)-\hat{\rho}^2(1)}{1-\hat{\rho}^2(1)} = -0.58$.
Since $\hat{R}(0) = 1.5$ we have $\hat{\sigma}^2 = \hat{R}(0)(1 - \hat{a}_1\hat{\rho}(1) - \hat{a}_2\hat{\rho}(2)) = 1.5(1 - 1.42 \cdot 0.9 + 0.58 \cdot 0.7) = 0.192$.

Checking causality of ARMA process

Exercise. Show that the ARMA process satisfying $x_t = x_{t-1} + 5cx_{t-2} - 5cx_{t-3} + \epsilon_t$ is not causal for all c.

We have to find roots of the polynomial

$$a(z) = 1 - z - 5cz^2 + 5cz^3.$$

Note that

$$a(z) = 1 - z - 5cz^2(1 - z) = (1 - z)(1 - 5cz^2).$$

Solving $a(z) = 0$, we can see that one root equals 1, which is not located outside the unit circle.
Thus, this process is not causal.

Seasonal ARIMA

Seasonal ARIMA [abbr. SARIMA] is an extension of the ARIMA model to stochastic processes with periodic behaviour.

A process x_t is a seasonal ARIMA$(p,d,q)(P,D,Q)^s$ process with period s if the differenced series $y_t = (1-B)^d(1-B^s)^D x_t$ is a causal seasonal ARMA process defined by

$$a(B)\tilde{a}(B^s)y_t = b(B)\tilde{b}(B^s)\varepsilon_t$$

where ϵ_t is a white noise with mean 0 and variance σ^2, polynomials $a(z) = 1 - a_1 z - a_2 z^2 - \ldots - a_p z^p$ and $b(z) = 1 + b_1 z + \ldots + b_q z^q$ have no common roots, polynomials $\tilde{a}(z) = 1 - \tilde{a}_1 z - \tilde{a}_2 z^2 - \ldots - \tilde{a}_P z^P$ and $\tilde{b}(z) = 1 + \tilde{b}_1 z + \ldots + \tilde{b}_Q z^Q$ have no common roots, $d, D \in \{0, 1, 2, \ldots\}$.

SARIMA$(1,0,1)(1,0,0)^{12}$

SARIMA models are often used to describe monthly time series, where the period is $s = 12$.

For example, the difference equation of SARIMA$(1,0,1)(1,0,0)^{12}$ is

$$(1 - a_1 B)(1 - \tilde{a}_1 B^{12}) x_t = (1 - b_1 B)\varepsilon_t$$

or, equivalently,

$$x_t - a_1 x_{t-1} - \tilde{a}_1 x_{t-12} + a_1 \tilde{a}_1 x_{t-13} = \varepsilon_t - b_1 \varepsilon_{t-1},$$

which looks like ARIMA(13,0,1) but has fewer parameters.

Forecasting SARIMA processes is completely analogous to forecasting ARIMA processes.

Learning with AI

Conceptualizing ARIMA

Prompt: "Explain the 'Integrated' part of an ARIMA(p,d,q) model. How does differencing a time series (where $d = 1$) help achieve stationarity, and what happens if we over-difference?"

SARIMA Complexity

Prompt: "Help me understand the notation for SARIMA$(1, 0, 1)(1, 1, 0)^{12}$. Walk me through what each parameter means in the context of monthly data with a yearly seasonal pattern."

Practice with Calculations

Task: Provide an AI with a specific ARIMA(1,1,0) equation and 3 data points (like the gold price example on slide 9). Ask it to generate the next 2 forecasts and explain the recursive steps to verify your own hand-calculations.

Homework

- Solve the past 2020 exam paper, which is provided separately on Learning Central.